Lecture #22: Beyond the ANOVA Procedure

STAT 200 – Introductory Statistics

Torin Quinlivan

Knox College

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Review: Procedure Requirements

Always, we have to make assumptions in determining the distribution of the test statistic. In many cases, you saw the assumptions. In some, they were hidden.

Review: Procedure Requirements

Test	**Assumption**
Z-test	Normality
t-test	Normality
Variance Test	Normality
Wilcoxon Test	Symmetry
Mann-Whitney Test	none
Binomial Test	none
Proportions Test	Normality
Chi-Square Goodness-of-Fit Test	Normality

ANOVA Requirements

For the analysis of variance procedure, there are two assumptions that we needed to make:

- Normality in each group.
- Equal variances across the groups.

ANOVA Requirements

These assumptions can be tested using the following tests:

- Shapiro-Wilk test (`shapiroTest`)
- Fligner-Killeen test (`fligner.test`)

The null hypothesis for each test is:

- H_0 : the data are from a Normal distribution.
- H_0 : The variances are equal. ($\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$)

Thus, small p-values indicate the requirement is not met by the data.

Example #1: Rice Yields

Does rice variety influence average yield amongst these four varieties?
First, because we are testing for independence between a numeric variable (yield) and a categorical variable (variety), we would like to use the analysis of variance procedure.
The hypotheses are:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$
$$H_A : \text{At least one of the means differs from the others.}$$

Example #1: Rice Yields

The following code loads the data and tests the assumptions:

```
library(KnoxStats)
rice = read.csv("http://rfs.kvasaheim.com/data/rice.csv")
attach(rice)

shapiroTest(yield ~ variety)
```

```
## Shapiro-Wilk Test
## Adjustment: Bonferroni correction (k = 4 )
##
##  Level   p.value
##      A 0.3148640
##      B 1.0000000
##      C 0.5065811
##      D 0.8381207
```

```
fligner.test(yield ~ variety)
```

```
##
##         Fligner-Killeen test of homogeneity of variances
##
## data:  yield by variety
## Fligner-Killeen:med chi-squared = 1.0026, df = 3, p-value =
## 0.8006
```

Example #1: Rice Yields

Incomplete Conclusion: We are asked to determine if the rice yield and the rice variety are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a Normal distribution in each group; and the variances are the same across the groups. Neither assumption is violated. The minimum p-value from the Shapiro-Wilk test is 0.0787, which is greater than $\alpha = 0.05$. The p-value from the Fligner-Killeen test, 0.8006, is also greater than our $\alpha = 0.05$. Because neither assumption is violated, we can use the analysis of variance procedure.
The small p-value of 0.00503 of the ANOVA procedure indicates that the two variables, yield and variety, are dependent. Not all varieties have the same average yield.

Beyond ANOVA

Note that the conclusion was *at least one mean differed from the others.* That is hardly helpful.
What we really want to know is "*which* variety is different?"
ANOVA cannot answer that question.

Tukey's HSD Test

However, there is a procedure that *can* answer it: **Tukey's Honestly Significant Difference Test**.
Its assumptions are the same as for ANOVA. So, if you use ANOVA, then you can use Tukey's HSD test.

Tukey's HSD Test

The code to perform Tukey's HSD test on our model is:

```
ricemod = aov(yield ~ variety)

TukeyHSD(ricemod)
```

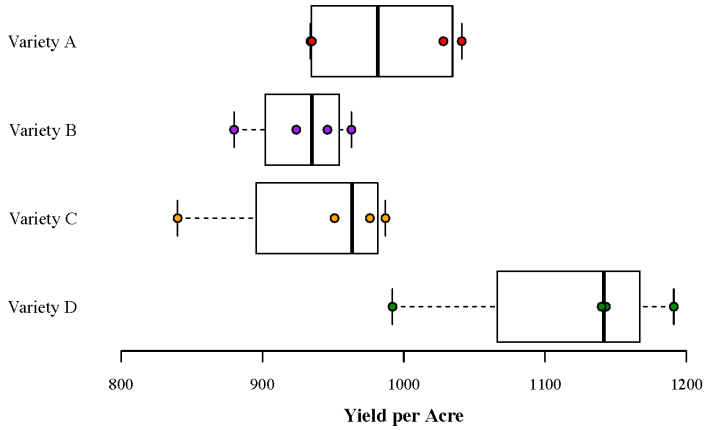
```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = yield ~ variety)
##
## $variety
##       diff         lwr      upr     p adj
## B-A -56.25 -191.592699  79.0927 0.6185496
## C-A -46.00 -181.342699  89.3427 0.7473470
## D-A 132.00   -3.342699 267.3427 0.0567296
## C-B  10.25 -125.092699 145.5927 0.9957690
## D-B 188.25   52.907301 323.5927 0.0066015
## D-C 178.00   42.657301 313.3427 0.0097522
```

Tukey's HSD Test

Find the p-values and interpret them for the null hypothesis "the difference in averages between these two levels is 0."
At our usual level of significance, we were only able to detect differences in average yield between Varieties B and D and between Varieties C and D.
Variety D has a higher average yield than Variety B by between 53 and 324.
Variety D has a higher average yield than Variety C by between 43 and 313.
We did not detect a difference between average yield between any other comparisons.

Tukey's HSD Test

These conclusions are hardly surprising in light of the data:



Kruskal-Wallis Test

If any of the assumptions necessary for ANOVA are violated, you should *not* use ANOVA. Like the t-test, there is a non-parametric alternative to ANOVA. It is called the **Kruskal-Wallis test**.

- It has the same requirements as the Mann-Whitney test.
- It is built into `R` (`kruskal.test`); its multiple comparisons test, `kruskal()`, is part of the `agricolae` package, which you have to install.
- Its output is simpler than that of ANOVA.
- It has its own multiple comparisons test, the Kruskal (multiple comparisons) test.
- The Kruskal test output is also simpler than that of Tukey's HSD.

Example #2: Citizen's Initiative

The citizen's initiative allows the people of the state to force a vote on a given issue. Do the different political cultures use the initiative at the same rates?
The hypotheses are:

$$H_0 : \mu_m = \mu_i = \mu_t$$
$$H_A : \text{At least one of the means differs from the others.}$$

Example #2: Citizen's Initiative

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a Normal distribution in each group.
- The data have the same variance across the groups.

Example #2: Citizen's Initiative

To check this using R, we run the following code:

```
shapiroTest(inituse ~ domPolCulture)
```

```
## Shapiro-Wilk Test
## Adjustment: Bonferroni correction (k = 3 )
##
##             Level      p.value
##        Moralistic 2.048182e-03
##   Individualistic 1.357651e-03
##  Traditionalistic 9.940607e-06
```

```
fligner.test(inituse ~ domPolCulture)
```

```
##
##         Fligner-Killeen test of homogeneity of variances
##
## data:  inituse by domPolCulture
## Fligner-Killeen:med chi-squared = 10.007, df = 2, p-value =
## 0.006715
```

Example #2: Citizen's Initiative

Because *at least one* requirement was not met, we should not use ANOVA.
We will use the Kruskal-Wallis test:

```
kruskal.test(inituse, domPolCulture)
```

```
##
##         Kruskal-Wallis rank sum test
##
## data:  inituse and domPolCulture
## Kruskal-Wallis chi-squared = 6.3238, df = 2, p-value = 0.04235
```

Because the p-value is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #2: Citizen's Initiative

To determine *which is different*, we use Kruskal's multiple comparisons test:

```
print(kruskal(inituse, domPolCulture))
```

```
## $statistics
##      Chisq Df    p.chisq  t.value      MSD
##   6.323785  2 0.04234553 2.010635 8.932211
##
## $parameters
##             test p.ajusted        name.t ntr alpha
##   Kruskal-Wallis      none domPolCulture   3  0.05
##
## $means
##                      inituse     rank       std  r Min Max Q25 Q50 Q75
## Individualistic   4.705882 24.91176  6.272606 17   0  18   0   0  10
## Moralistic       14.764706 32.32353 19.533719 17   0  60   0   8  16
## Traditionalistic  2.764706 20.76471  5.984662 17   0  23   0   0   2
##
## $comparison
## NULL
##
## $groups
##                     inituse groups
## Moralistic       32.32353      a
## Individualistic  24.91176     ab
## Traditionalistic 20.76471      b
##
## attr(,"class")
## [1] "group"
```

Example #2: Citizen's Initiative

Treatments with the same letter are not significantly different.
From this, we know we detected a difference in average initiative use between the moralistic states and the traditionalistic states. No other differences were detected.

Example #2: Citizen's Initiative

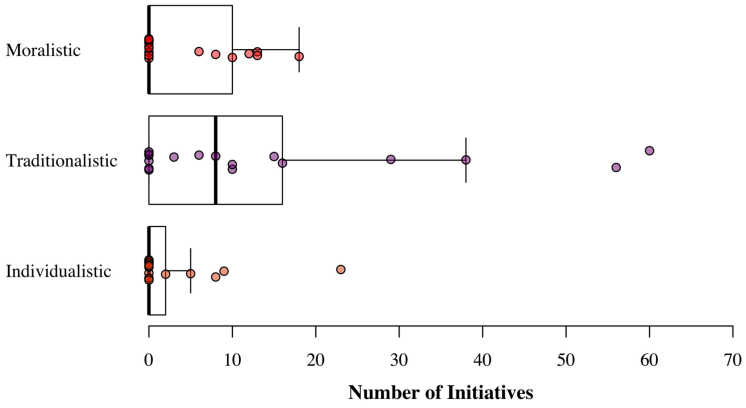
Conclusion:
We are asked to determine if the initiative use and the political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a Normal distribution in each group; and the variances are the same across the groups. Both assumptions are violated. The maximum p-value from the Shapiro-Wilk test is 0.00204, which is much less than $\alpha = 0.05$. The p-value from the Fligner-Killeen test, 0.0067, is also less than our $\alpha = 0.05$. Because both assumptions are violated, we should not use the analysis of variance procedure. We must use the Kruskal-Wallis test.

Example #2: Citizen's Initiative

Conclusion:
The small p-value of 0.0424 of the Kruskal-Wallis test indicates that the two variables, initiative use and political culture are dependent. Not all political cultures use the initiative process equally. According to the Kruskal multiple comparisons test, we can conclude that states with a moralistic political culture tend to use the initiative more frequently than states with a traditionalistic culture. No other comparisons were significant.

Example #2: Citizen's Initiative

Again, these conclusions make sense when seeing the graphic:



Example #3: Violent Crime

Does the 2000 violent crime rate significantly vary across the four census regions?
The hypotheses are:

$$H_0 : \mu_N = \mu_S = \mu_M = \mu_W$$
$$H_A : \text{At least one of the means differs from the others.}$$

Example #3: Violent Crime

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a Normal distribution in each group.
- The data have the same variance across the groups.

Example #3: Violent Crime

To check this using R, we run the following code:

```
shapiroTest(vcrime00 ~ census4)
```

```
## Shapiro-Wilk Test
## Adjustment: Bonferroni correction (k = 4 )
##
##      Level   p.value
##  Northeast 1.0000000
##    Midwest 1.0000000
##      South 0.2251474
##       West 1.0000000
```

```
fligner.test(vcrime00 ~ census4)
```

```
##
##         Fligner-Killeen test of homogeneity of variances
##
## data:  vcrime00 by census4
## Fligner-Killeen:med chi-squared = 6.0913, df = 3, p-value =
## 0.1073
```

Example #3: Violent Crime

Because no requirement was violated, we can use ANOVA. Here is the code:

```
cmod1 = aov(vcrime00 ~ census4)
summary(cmod1)
```

```
##             Df  Sum Sq Mean Sq F value  Pr(>F)
## census4      3  689569  229856   4.855 0.00505 **
## Residuals   47 2225268   47346
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Because the p-value of 0.00505 is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Again, which one?

Example #3: Violent Crime

To determine which is different, we use the Tukey HSD test.

```
TukeyHSD(cmod1)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = vcrime00 ~ census4)
##
## $census4
##                          diff        lwr       upr     p adj
## Northeast-Midwest  -55.21111 -310.76041 200.33819 0.9389452
## South-Midwest      196.87647  -21.62826 415.38120 0.0910331
## West-Midwest       -74.66154 -306.65974 157.33667 0.8266966
## South-Northeast    252.08758   13.18663 490.98853 0.0350137
## West-Northeast     -19.45043 -270.75207 231.85122 0.9968573
## West-South        -271.53801 -485.05941 -58.01661 0.0075817
```

From this, we know we detected a difference in average violent crime rate between the South and Northeast states and between the South and West states, where the South is significantly higher than both the Northeast and the West.

Example #3: Violent Crime

Conclusion:
We are asked to determine if the violent crime rate and the census region are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a Normal distribution in each group; and the variances are the same across the groups. Neither assumption is violated. The minimum p-value from the Shapiro-Wilk test is 0.2251, which is greater than $\alpha = 0.05$. The p-value from the Fligner-Killeen test, 0.1073, is also greater than our $\alpha = 0.05$. Because neither assumption is violated, we can – and should – use the analysis of variance procedure.

Example #3: Violent Crime

Conclusion:
The small p-value of 0.00505 of the ANOVA test indicates that the two variables, violent crime rate and census region are dependent. Not all census regions have the same average violent crime rate. According to Tukey's HSD test, we can conclude that southern states have a significantly higher average violent crime rate than the Northeastern states and the Western states. No other comparisons were significant.

Example #4: Education

Does the 2000 weighted average educational achievement (WAEA) significantly vary across the three political cultures?
The hypotheses are:

H_0 : All the means are the same.

H_A : At least one of the means differs from the others.

Example #4: Education

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a Normal distribution in each group.
- The data have the same variance across the groups.

Example #4: Education

To check this using R, we run the following code:

```
shapiroTest(waea00 ~ domPolCulture)
```

```
## Shapiro-Wilk Test
## Adjustment: Bonferroni correction (k = 3 )
##
##             Level   p.value
##        Moralistic 0.6071567
##   Individualistic 1.0000000
##  Traditionalistic 1.0000000
```

```
fligner.test(waea00 ~ domPolCulture)
```

```
##
##         Fligner-Killeen test of homogeneity of variances
##
## data:  waea00 by domPolCulture
## Fligner-Killeen:med chi-squared = 2.8698, df = 2, p-value =
## 0.2381
```

Example #4: Education

Because no requirement was violated, we can use ANOVA. Here is the code:

```
waeamod = aov(waea00 ~ domPolCulture)
summary(waeamod)
```

```
##               Df Sum Sq Mean Sq F value   Pr(>F)
## domPolCulture  2  276.0  138.02   17.19 2.35e-06 ***
## Residuals     48  385.4    8.03
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Because the p-value is much less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #4: Education

To determine which is different, we use the Tukey HSD test.

```
TukeyHSD(waeamod)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = waea00 ~ domPolCulture)
##
## $domPolCulture
##                                       diff        lwr       upr
## Moralistic-Individualistic         1.725882 -0.6247889  4.076554
## Traditionalistic-Individualistic -3.840588 -6.1912595 -1.489917
## Traditionalistic-Moralistic      -5.566471 -7.9171418 -3.215799
##                                       p adj
## Moralistic-Individualistic        0.1885858
## Traditionalistic-Individualistic 0.0007316
## Traditionalistic-Moralistic      0.0000019
```

From this, we know we detected a difference in average WAEA rate between the traditionalist states and both of the other two types. In both cases, the traditionalistic states tended to have lower WAEA. No other comparisons were statistically significant.

Example #4: Education

Conclusion:
We are asked to determine if WAEA and the dominant political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a Normal distribution in each group; and the variances are the same across the groups. Neither assumption is violated. The minimum p-value from the Shapiro-Wilk test is 0.6072, which is greater than $\alpha = 0.05$. The p-value from the Fligner-Killeen test, 0.2381, is also greater than our $\alpha = 0.05$. Because neither assumption is violated, we can – and should – use the analysis of variance procedure.

Example #4: Education

Conclusion:
The small p-value of 2.35×10^{-6} of the ANOVA test indicates that the two variables, WAEA and dominant political culture are dependent. According to Tukey's HSD test, we can conclude that traditionalistic states have a significantly lower average WAEA than the moralistic and individualistic states. No other comparisons were significant.

Example #5: Average Wealth

Does the 2000 GSP per capita significantly vary across the three political cultures? The hypotheses are:

$$H_0 : \mu_I = \mu_M = \mu_T$$
$$H_A : \text{At least one of the means differs from the others.}$$

Example #5: Average Wealth

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a Normal distribution in each group.
- The data have the same variance across the groups.

Example #5: Average Wealth

To check this using R, we run the following code:

```
shapiroTest(gspcap00 ~ domPolCulture)
```

```
## Shapiro-Wilk Test
## Adjustment: Bonferroni correction (k = 3 )
##
##             Level      p.value
##        Moralistic 7.771441e-01
##   Individualistic 7.609321e-03
##  Traditionalistic 3.239349e-05
```

```
fligner.test(gspcap00 ~ domPolCulture)
```

```
##
##         Fligner-Killeen test of homogeneity of variances
##
## data:  gspcap00 by domPolCulture
## Fligner-Killeen:med chi-squared = 0.77112, df = 2, p-value =
## 0.6801
```

Example #5: Average Wealth

Because at least one requirement was violated, we should not use ANOVA. We need to use the Kruskal-Wallis test. Here is the code:

```
kruskal.test(gspcap00 ~ domPolCulture)
```

```
##
##         Kruskal-Wallis rank sum test
##
## data:  gspcap00 by domPolCulture
## Kruskal-Wallis chi-squared = 13.752, df = 2, p-value = 0.001032
```

Because the p-value of 0.0010 is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #5: Average Wealth

To determine *which is different*, we use Kruskal's multiple comparisons test:

```
print(kruskal(gspcap00, domPolCulture))
```

```
## $statistics
##      Chisq Df     p.chisq  t.value      MSD
##   13.75193  2 0.001032301 2.010635 8.909261
##
## $parameters
##            test p.ajusted        name.t ntr alpha
##   Kruskal-Wallis      none domPolCulture   3  0.05
##
## $means
##                   gspcap00     rank       std  r      Min      Max
## Individualistic  39160.74 36.35294  8429.325 17 31213.49 65117.14
## Moralistic       32676.14 23.82353  4466.155 17 24059.87 39372.47
## Traditionalistic 32455.14 17.82353 12116.530 17 22636.06 76483.42
##                       Q25      Q50      Q75
## Individualistic  32644.85 37548.27 42483.13
## Moralistic       29760.26 31445.69 37002.88
## Traditionalistic 26831.35 29509.63 33909.05
##
## $comparison
## NULL
##
## $groups
##                   gspcap00 groups
## Individualistic  36.35294      a
## Moralistic       23.82353      b
## Traditionalistic 17.82353      b
##
## attr(,"class")
## [1] "group"
```

Treatments with the same letter are not significantly different.

Example #5: Average Wealth

From this, we know we detected a difference in average GSP per capita between the individualistic states and both of the other two types. In both cases, the individualistic states tended to have higher GSP per capita. No other comparisons were statistically significant.

Example #5: Average Wealth

Conclusion:
We are asked to determine if the average GSP per capita and the dominant political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a Normal distribution in each group; and the variances are the same across the groups. The Normality assumption is violated. The minimum p-value from the Shapiro-Wilk test is 0.000033, which is much less than $\alpha = 0.05$. Because an assumption is violated, we need to use the Kruskal-Wallis test.

Example #5: Average Wealth

Conclusion:
The small p-value of 0.0010 of the Kruskal-Wallis test indicates that the two variables, GSP per capita and dominant political culture are dependent. According to the Kruskal multiple comparison test, we can conclude that individualist states have a significantly higher GSP per capita than the moralistic and traditionalistic. No other comparisons were significant.

Example #6: Legislature Professionalism

Does the professional level of the state's legislature significantly vary across the three political cultures?
The hypotheses are:

$$H_0 : \mu_I = \mu_M = \mu_T$$
$$H_A : \text{At least one of the means differs from the others.}$$

Example #6: Legislature Professionalism

Because we are comparing multiple means, we would like to use the ANOVA procedure. It has two assumptions:

- The data come from a Normal distribution in each group.
- The data have the same variance across the groups.

Example #6: Legislature Professionalism

To check this using R, we run the following code:

```
shapiroTest(profleg ~ domPolCulture)
```

```
## Shapiro-Wilk Test
## Adjustment: Bonferroni correction (k = 3 )
##
##             Level     p.value
##        Moralistic 0.002017848
##   Individualistic 0.033975358
##  Traditionalistic 1.000000000
```

```
fligner.test(profleg ~ domPolCulture)
```

```
##
##         Fligner-Killeen test of homogeneity of variances
##
## data:  profleg by domPolCulture
## Fligner-Killeen:med chi-squared = 5.1794, df = 2, p-value =
## 0.07504
```

Example #6: Legislature Professionalism

Because at least one requirement was violated, we should not use ANOVA. We need to use the Kruskal-Wallis test. Here is the code:

```
kruskal.test(profleg ~ domPolCulture)
```

```
##
##         Kruskal-Wallis rank sum test
##
## data:  profleg by domPolCulture
## Kruskal-Wallis chi-squared = 6.5777, df = 2, p-value = 0.0373
```

Because the p-value of 0.0373 is less than our usual $\alpha = 0.05$, we reject the null hypothesis. We can conclude that at least one mean differs from the others. Which one?

Example #6: Legislature Professionalism

To determine *which is different*, we use Kruskal's multiple comparisons test:

```
print(kruskal(profleg, domPolCulture))
```

```
## $statistics
##      Chisq Df    p.chisq
##   6.577695  2 0.03729682
##
## $parameters
##            test p.ajusted        name.t ntr alpha
##   Kruskal-Wallis      none domPolCulture   3  0.05
##
## $means
##                     profleg     rank        std  r  Min  Max  Q25   Q50
## Individualistic  0.2764706 32.82353 0.15519911 17 0.06 0.66 0.19 0.260
## Moralistic       0.2152941 22.35294 0.17507561 17 0.04 0.65 0.11 0.160
## Traditionalistic 0.1712500 21.06250 0.05264662 16 0.10 0.26 0.13 0.165
##                     Q75
## Individualistic  0.3100
## Moralistic       0.2300
## Traditionalistic 0.2025
##
## $comparison
## NULL
##
## $groups
##                   profleg groups
## Individualistic  32.82353      a
## Moralistic       22.35294      b
## Traditionalistic 21.06250      b
##
## attr(,"class")
## [1] "group"
```

Treatments with the same letter are not significantly different.

Example #6: Legislature Professionalism

From this, we know we detected a difference in average level of legislative professionalism between the individualistic states and both of the other two types. In both cases, the individualistic states tended to have higher levels of legislative professionalism. No other comparisons were statistically significant.

Example #6: Legislature Professionalism

Conclusion:
We are asked to determine if the level of legislative professionalism and the dominant political culture are independent. To do this, we would prefer to use the analysis of variance procedure, because it is the most powerful of the available tests. It has two requirements: The data are from a Normal distribution in each group; and the variances are the same across the groups. The Normality assumption is violated. The minimum p-value from the Shapiro-Wilk test is 0.0020, which is much less than $\alpha = 0.05$. Because an assumption is violated, we need to use the Kruskal-Wallis test.

Example #6: Legislature Professionalism

Conclusion:
The small p-value of 0.0373 of the Kruskal-Wallis test indicates that the two variables, legislative professionalism and dominant political culture, are dependent. According to Kruskal's multiple comparisons test, we can conclude that individualistic states tend to have a significantly higher level of legislative professionalism than the moralistic and traditionalistic states. No other comparisons were significant.